Registered report: Mapping Synesthetic Sequences in Space: Improving Consistency Test by Harnessing Cartography Tools.

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Abstract

Sequence-space synesthesia is a perceptual condition in which ordinal sequences (e.g., numbers, days of the week, or months of the year) are experienced as occupying specific spatial locations. This perceptual condition is identified using self-reports (e.g. questionnaires) and behavioural consistency test methods. Existing consistency methods measure the spatial distance between responses to repeated stimuli, but they ignore the ordinal and geometric properties of spatial-forms. This registered report consists of two stages. In *phase I*, we optimise SSS diagnostics from consistency test by extracting novel tests at the spatial-form level. To do this, we first evaluate classification performance of old and new methods in a large (N = 685) aggregated sample from four available and independent datasets. We then harness a geographic toolbox to extract metrics from the spatial-forms level. Finally, receiver operating characteristic analyses are conducted to assess the discriminative performances of each method. The results show that permuted topological validity of spatial-forms and the perimeter between stimuli performs equally well in detecting SSS. In *phase II*, we will examine predictive validity of the selected methods on an independent dataset that has yet to be collected. Findings from *phase I* highlight the relevance of topological principles for sequence-space representations and aim to optimise the ability of consistency tests to detect SSS.

Keywords: Sequence space synaesthesia, Synaesthesia/synesthesia, consistency test, space, time, numbers

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1 Introduction

Sequence-Space Synaesthesia (SSS), or visuo-spatial forms, is the phenomenon where people visualize ordered sequences in particular spatial positions. For example, numbers, weekdays or months (synesthetic *inducers*) are represented as arranged into specific spatial positions in space (synesthetic *concurrents*). The synaesthetic spatial-forms can be complex and have variable geometric forms. These geometric forms might be different for each category (i.e. number-forms weekdays and months), with forms such as ellipses, zig-zags or curbed lines (Flournoy, 1893; Galton, 1880). However, the spatial-forms of individuals with SSS are consistent and automatic (Deroy & Spence, 2013; Seron et al., 1992). Existing methods for classifying SSS and controls based on consistency tests have some limitations (Baron-Cohen et al., 1987; Brang et al., 2010; Eagleman et al., 2007; Rothen et al., 2016; Sagiv et al., 2006; Ward et al., 2018; Ward & Simner, 2022). The aim of the present exploratory phase 1 study is to optimize consistency methods from aggregated and already available datasets. Then, in confirmatory phase 2 registered report study, we aim to test and compare the optimal methods on data that has yet to be collected.

SSS frequently co-occurs with other subtypes of synaesthesia and as for other subtypes of synaesthesia it is idiosyncratic. Grapheme-colour synaesthesia, where a grapheme *inducer* triggers a *concurrent* colour (i.e. “A” seen as red), co-occurs at 71% to 76% with SSS (Sagiv et al., 2006). Nevertheless, across different synaesthetic subtypes, self-reported SSS tend to cluster together (Ward & Simner, 2022), indicating a degree of internal homogeneity and justifying the study of it as a distinct phenomenon. Spatial-forms are idiosyncratic, which results in considerable heterogeneity in how this phenomenon is manifested across individuals. One source of heterogeneity is given by dimensionality: some SSS experiences can involve three-dimensional (3D) and two-dimensional (2D) arrangements (Eagleman, 2009; Price & Pearson, 2013). Another source of heterogeneity is the reference frame, for example, the spatial-forms might take place in an external space around the body (*i.e.* projector) or in an egocentric internal space (*i.e.* associator) (Dixon et al., 2004; Smilek et al., 2007). Further variability can be explained by temporal-spatial properties and mental-manipulation of spatial-forms such as "zooming" in and out or rotating or shifting perspectives at will (Gould et al., 2014; Jarick et al., 2009). Lastly, the shape, complexity and layout of the spatial-forms are also heterogeneous such as forming for example ovals, lines, zig-zags or loops. With some recurring shapes being more frequent across SSS, such as ovals for months (Eagleman, 2009).

1.1 Consistency tests

Individuals with SSS are usually identified by the means of phenomenological reports such as questionnaires and consistency tests. Questionnaires rely on self-report of the consistency, automaticity, directionality and developmental onset of spatial associations. Consistency tests rely on more objective behavioural tasks used to evaluate how stable synesthetic experiences are over time (Rothen et al., 2013, 2016).

In the SSS consistency task, participants complete multiple rounds of the same stimuli (e.g., digits, days of the week, and months of the year) and are asked to indicate a spatial location for each stimulus by clicking on the computer screen. Hence, consistency tasks are designed to assess how similar participant's responses are across repetitions (i.e. *inducer-concurrent* associations). An individual is identified as a synaesthete based on the variability between repeated responses to the same inducer or stimuli: less variable responses being an indicator of the consistent responses characteristic of synaesthesia. This because synaesthetes can respond from their perceptual experience while non-synesthetes can only be consistent by relying on a memorization strategy. Consistency tests have proven effective for colour-grapheme synaesthesia. Measures of individual consistency can be derived using colour pickers while repeatedly presenting the same inducer (i.e. "A"). The quantification of the distance between concurrents from the same inducers is used to discriminate consistent synaesthetes from controls (Rothen et al., 2013). A similar rationale to that used for colour-grapheme synaesthesia has been applied to characterize SSS from consistency tasks.

Brang et al. (2010) for example evaluated consistency as the distance between repeated responses to the same inducer or stimuli (e.g. January and January) relative to adjacent stimuli (e.g. February). A response was defined as consistent if it fell within 1.96 *z-scores*. However, this criterion was noted to be potentially too conservative, as it identified synaesthesia in 4 of 81 self-reported synaesthetes. Geometrical features such as area and perimeter between the coordinates of repeated inducers have further been used as a measure of the distance between same inducers, hence as a metric for consistency. Less consistent responses lead to smaller triangle perimeters and area (Rothen et al., 2016). A cut-off has been proposed whereby individuals are classified as SSS if their average response area is smaller than 0.203% of the total screen area (see also Ward et al., 2018).

In contrast to the consistency tests for colour-grapheme synaesthesia, performances of consistency test for SSS at the suggested cut-off are less performant. For example, existing consistency tests for Colour-grapheme lead to an Area Under the Curve of 93%, which corresponds to the probability of the test to correctly classify synesthetes from non synaesthetes (Sensitivity = 90%, Specificity = 94%, see Rothen et al. (2013)). In contrast, consistency tests for SSS lead to an Area Under the Curve of 76% (Sensitivity = 88%, Specificity = 77%, see Rothen et al. (2016)).

1.2 Limitations of current methods of consistency tests

A first limitation of current SSS consistency tests is that participants that respond with the same position, for example when not knowing the response, will have artificially high consistencies with those metrics (Rothen et al., 2016). Indeed, exclusion via visual inspection of these responses lead to improved classification performances using the area test (i.e. Sensitivity = 87%; Specificity = 81%, Area Under the Curve = 85%, see Rothen et al. (2016)). A second limitation is that consistency methods may favour certain spatial-forms(Ward et al., 2018) in particular those with linear or highly regular spatial layouts, i.e. elliptical patterns for months(Brang et al., 2010; Eagleman, 2009; Flournoy, 1893). This has led to alternative approaches such as to add the standard deviation of responses, questionnaire cut-off (Ward et al., 2018) or permutation-based comparison of individual responses to chance levels for colour-grapheme (Root et al., 2021) and SSS (Ward, 2022). Third, these consistency tests evaluate the variability of responses to the same stimuli, therefore the ordinality between stimuli is not taken into account. Synaesthetic spatial-forms follow ordinal rules (e.g. Monday → Tuesday → Wednesday). In the domain of numerical cognition, ordinality has been identified as an important information when processing sequences (Lyons & Beilock, 2013). Some studies of SSS have systematically investigated ordinality by investigating metrics of adjacent inducer-concurrent pairs such as distances (Brang et al., 2010) or angles (Eagleman, 2009). However, ordinality of the concurrent spatial-form might be particularly relevant considering all the stimuli of a certain category (i.e. weekdays or months).

Another limitation of consistency tests for SSS concerns the number category. Indeed, spatial representations for numbers are also phenomenological in the general population. It has been theorized the existence of a mental number line, where numbers are spatially represented from small-left to large-right(Dehaene et al., 1993; Hubbard et al., 2005). Experimentally, the Spatial Numerical Association of Response Code, in short SNARC effect, finds that smaller numbers are responded faster with the left hand and larger number with the right hand(Dehaene et al., 1993). Hence controls could have an internalized representation for numbers in space. It is however important to note that the SNARC is not reliable on an individual level (Cipora et al., 2019; Ramos et al., 2026) and that the SNARC effect might also arise from a response rather than a representational level (Basso Moro et al., 2018; Keus & Schwarz, 2005). Nevertheless this phenomenon might lead to consistent responses for numbers also by controls.

Here, we aim at optimizing the identification of SSS from consistency tasks using geometrical characteristics of spatial-forms from ordered stimuli or inducers.

1.3 Present study

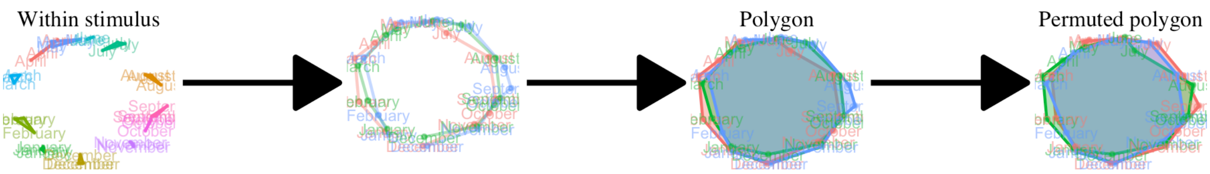
This registered consists of two phases, *phase I* is an exploratory study to find an optimal consistency test to classify SSS and controls, on available aggregated datasets. *Phase II* is a confirmatory study of the most optimal consistency tests on data that has yet to be collected.

In the present *Phase I*, we merge four previously available datasets using the same task on both SSS and control groups (Ward, 2022). First, we aimed to reproduce established consistency test methods based on stimulus-level consistency metrics, such as area and perimeter. Second, we explored a new approach comparing geometrical features across repetitions of the same categories (weekday, month and numbers), such as segments and polygons. These novel tests are designed to take advantage of the sequentiality or ordinality between the responses and the geometrical properties at the spatial-form level, see Figure 1. All the consistency tests are then optimized using ROC analyses based on self-reported classification self-reported SSS and control groups.

In *Phase II*, we will assess whether the tests identified in the present study generalize and are validated on an independent dataset that is yet to be acquired (Sachdeva et al. (2024), Stage 1 in-principle acceptance, see also https://osf.io/6wn4m).

Figure 1

Design for novel method for consistency test from between trial varibility to Segment and polygons formed by the space-forms using geography tools.



Note. Example of a participant's months space-form illustrating the novel method's design.

2 Methods - phase I

Code and data for "one-click" reproducibility of this manuscript including the following analyses are available on git-hub (https://github.com/remLach/SpaceSequenceSynDiagnostic). All the analyses are conducted in R, using the following packages 'pROC' v. 1.19.0.1 (Robin et al., 2011), 'sf' v. 1.0.21 (Pebesma & Bivand, 2023) and for visualization 'ggplot2' v. 4.0.1 (Wickham, 2016), 'ggVennDiagram' v. 1.5.6 (Gao & Dusa, 2025).

2.1 Datasets and Participants

To base the optimization of the consistency test on the largest possible sample we looked for accessible datasets that used the same consistency tasks on groups of synesthetes and controls. We found four datasets that met these criteria. Two datasets were collected in laboratory settings Rothen et al. (2016) (see also https://osf.io/6hq94) and Van Petersen et al. (2020) (see also https://data.ru.nl/collections/di/dcc/DSC_2018.00019_653). The two others came from on-line testing Ward (2022) (see also https://osf.io/nu5v4) and an additional dataset was provided by private communications with Jamie Ward. To match the other datasets, stimuli of the weekdays and months categories were translated from Dutch to English in the dataset from (Van Petersen et al., 2020). For the number category we included only digits from 0 to 9 that where presented across all datasets (i.e. excluding the stimuli "50" and "100" for numbers in Van Petersen et al. (2020)).

From the merged datasets, we kept 685 from the total 689 participants. First, we excluded 0.02 % empty trials (i.e. skipped responses) including trials flagged for having the same x or y coordinates across categories and repetitions, causing the depletion (n = 2) participants (as in Rothen et al., 2016; Ward et al., 2018). (n = 2) participants were excluded for having less than 4 coordinate points since polygons need at least four coordinate. Note therefore, that 31 participants of the final sample did not have responses in all three categories b< repetition cases. From the final sample of N = 685, 396 were self-reported synaesthetes and 289 self-reported controls, see Table 1 and Table 2.

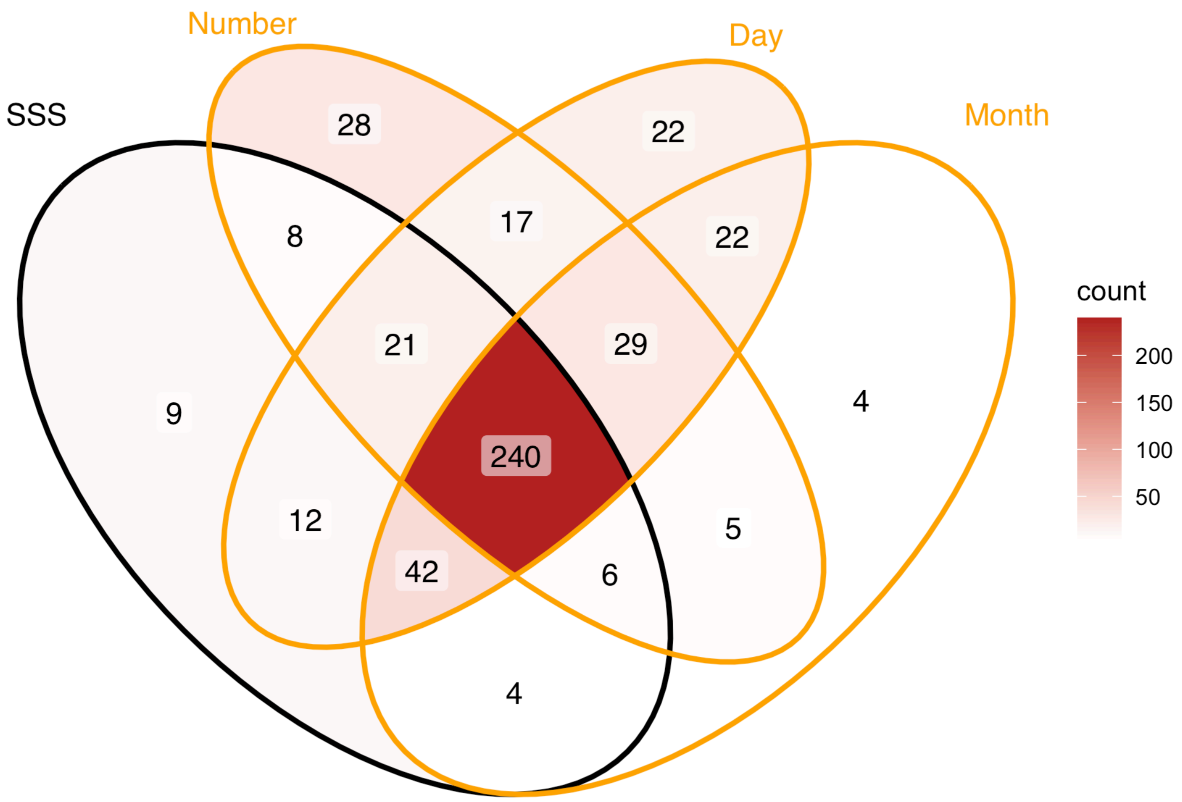
Regarding the synaesthetes profiles, detailed participant information such as questionnaire responses were only available from the two datasets from the data by Jamie Ward (i.e. a majority of 573 participants). We described the self-reported profiles for the stimulus categories used in the consistency test (i.e. number, weekdays and month), see Figure 2.

2.2 Procedure

For the consistency task, stimuli were presented randomly and sequentially centrally on the screen. Participant were instructed to click on the screen position where they visualize

Figure 2

Venn diagram of the types of self-reported SSS



Note. Only data from Pr. Ward is represented here. SSS = Sequence-space synaesthesia

Table 1

Synesthetes

Synaesthetes	Initial N	Final N	Age
Rothen et al. (2016)	33	32	23.1
Van Petersen et al. (2020)	23	22	23.2
Ward (2022)	252	252	37.2
Ward 2	90	90	NA
Total:	398	396	

Note. Synesthetes. Demographics were not provided in the dataset Ward 2. Before: before participant exclusion

Table 2

Controls

Controls	Initial N	Final N	Age
Rothen et al. (2016)	37	37	28.2
Van Petersen et al. (2020)	21	21	21.6
Ward (2022)	215	213	19.9
Ward 2	18	18	NA
Total	291	289	

Note. Controls. Demographics were not provided in the dataset Ward 2. Before: before participant exclusion

them. In Van Petersen et al. (2020) and Rothen et al. (2016) the participants were allowed to skip responses.

Stimuli included here were 7 weekdays (Monday to Sunday), 12 months (January to December) and 9 numbers (0 to 9). For all the data from Jamie Ward the stimuli were presented in randomized order with the constraint that no stimulus was repeated until all unique stimuli (N = 29) had been presented once. The median display resolution was 1440 X

768, with a maximum of 2560 X 2025 and a minimum of 308 X 149.

2.3 Analyses

We first reproduced existing methods and test their classification performances (i.e. ability to class controls as controls and SSS as SSS) using Receiver Operating Characteristics on the aggregated dataset. We then compared these classification performances to the ones of novel methods, see Figure 1.

2.3.1 Reproduced consistency tests

We reproduced two existing consistency methods used to quantify the spatial variability of responses between repetitions of the same stimuli (Rothen et al., 2016; Van Petersen et al., 2020; Ward, 2022). Synaesethes relying on perceptual spatial representations should have more consistent responses, hence less variable (i.e. closer) across the same stimuli across repetitions than controls. Hence, variability was estimated by measuring the distance between responses to the same stimuli (i.e. between repetitions to the same stimuli, see first panel of Figure 1).

Formally the area (see Equation 1) and perimeter (see Equation 2) were calculated from the triangle (i.e. given three repetitions) formed by the x and y coordinates of the three repetitions of each stimuli (i.e. between the coordinates *(x1, y1), (x2, y2), (x3, y3)*).

$$Area = \frac{|x1y2 + x2y3 + x3y1 - x1y3 - x2y1 - x3y2|}{2} \tag{1}$$

$$\begin{aligned} Perimeter = &\sqrt{(x2 - x1)^2 + (y2 - y1)^2} + \\ &\sqrt{(x3 - x2)^2 + (y3 - y2)^2} + \\ &\sqrt{(x1 - x3)^2 + (y1 - y3)^2} \end{aligned} \tag{2}$$

Each stimulus triangle area was then averaged by participants. The area metric is in % of the screen area to be able to compare with the consistency across studies using different screen sizes. To account for screen size and spread variability across the datasets and participants, area and perimeter were also calculated on individually normalized *x* and *y* coordinates (*z-score*).

For example, if a participant's set of *x,y* coordinates for Monday are close to each other, then the triangle area and perimeter will be smaller (i.e. more consistent and characteristics of synaesthetes). The recognized limit of this method, is that controls clicking on the same position can achieve artificially high consistencies, see for example the first four rows of Figure 3.

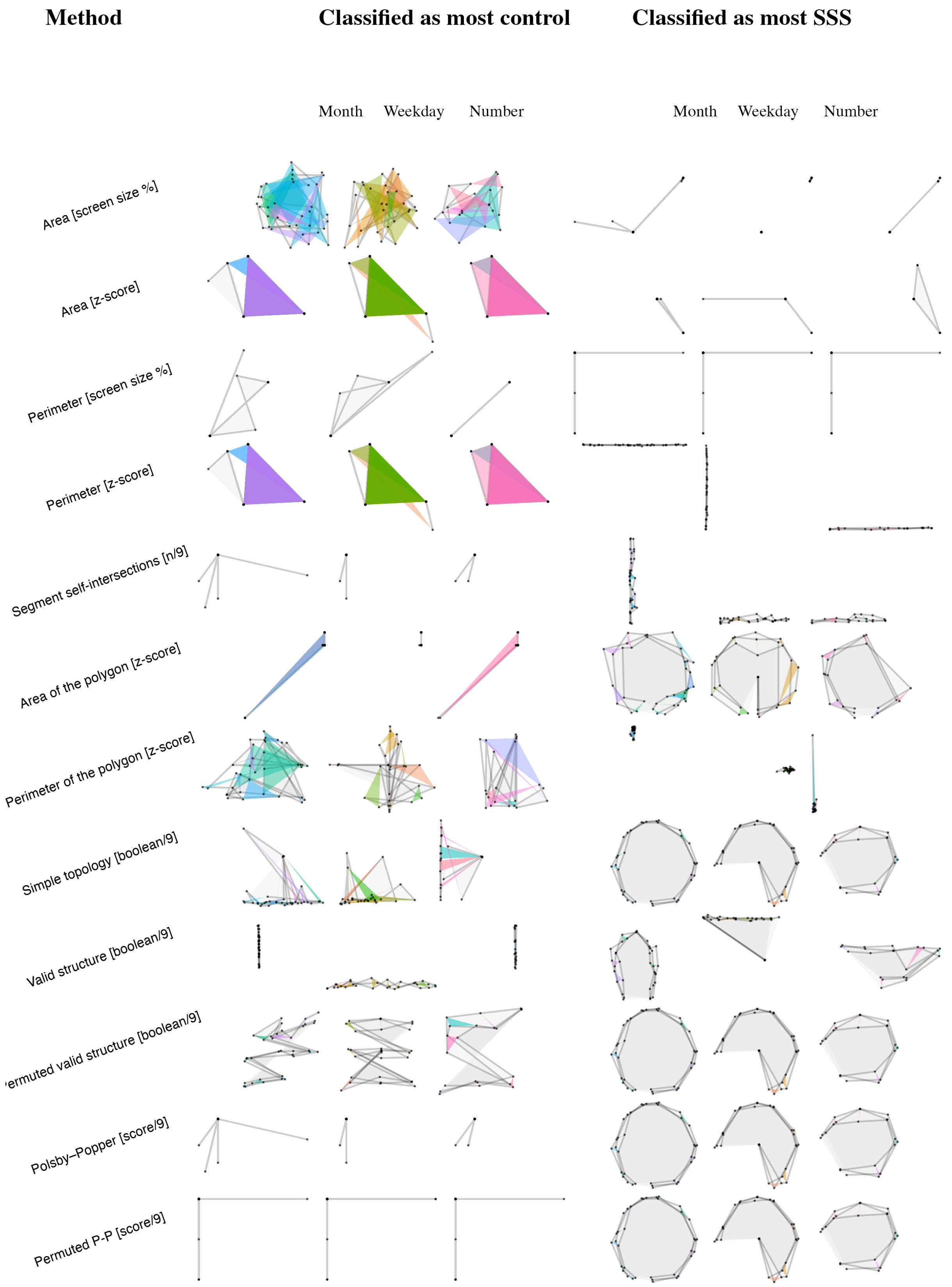
2.3.2 Consistency tests on the **space-form level**

For the new methods for calculating consistency, we extracted geometrical features at the spatial-form level for each repetition of sequentially ordered stimulus (i.e. for numbers: 0 →1→2→3→4→5→6→7→8→9). Each participant repeated each stimuli three times, leading to three spatial-form per categories (numbers, weekdays and months) that is a total nine space-forms per participants. Each space-form is evaluated individually using different methods. Note that the stimuli forming the space-forms are selected by the first time or repetition each one is presented during the experiment (i.e. Monday (repetition 1) → Tuesday (repetition 1) → Wednesday (repetition 1), ect). The rationale was that spatial-forms from SSS should have characteristic geometrical properties that are distinct than control's ones. Later on we extract the same geometrical properties on segments and polygons where the repetitions labels are permuted within each categories (i.e. Monday (repetition 2) → Tuesday (repetition 1) → Wednesday (repetition 3), ect). The rationale here is that spatial-forms should have the same geometrical properties independently from coordinate repetition order of the responses during the experiment.

Each space-form can be considered as a geometrical segment (i.e. open geometrical form) or polygon (i.e. closed geometrical form) laying on 2D Cartesian plane, similarly as originally described in Galton (1880), see Figure 1. Hence, we harnessed a geospatial analysis, the 'sf' package version 1.0.21 (Pebesma & Bivand, 2023) to extract polygons and

Figure 3

Example of extremes at the consistency tests



Note. Plotted are space-forms from the participants that are the extreme of the consistency test's value geometry-based features from participant's 2D (x,y) coordinate responses. This package allows then, for example, to build and analyse segments or polygons and then extract multiple geometrical descriptors or features. Informed by the ordinality of the stimulus, we defined segments and polygon by categories and repetitions.

For example, Figure 1, shows how we can build segments and polygons for months. This polygonal space-form has geometrical properties that distinguishes it from random or pseudo-random responses. For example, each of the three segments do not repeat. Figure 1 also illustrates the principle of permuting the repetitions each of the three polygons are re-drawn using the stimuli from different repetitions. For consistent space-forms the permutations should lead to polygons with the same geometrical properties.

Segment self-intersections. The first consistency test extracted from the segments is self-intersection. Each space-form considered as ordered segments can lead to different number of self-intersections. We reasoned that responses from controls should be less structured leading to self-intersections while for SSS the chances of self-intersections are lower. In addition it addresses the issue of participants giving the same responses, since with this test these responses would lead to a large number of self-intersections and hence classification to control.

We considered the spatial-forms as a segment constituted by the lines connecting consecutive stimuli separately for each category and repetitions (i.e. Monday (repetition 1) → Tuesday (repetition 1) → Wednesday (repetition 1), etc.) and then counted the number of times each segment self-intersect. The number of self-intersections were computed separately for each of the repetitions and categories and averaged per participants.

For example, Figure 4 illustrate a participant with the sum of self-intersections for each categories (columns) and repetitions (rows). For the self-intersection test, this participants value would be $((4+1+2)+(2+1+0)+(3+0+0))/9 = 1.44$.

Topological validity and simplicity. We assessed topological propetries using geometric validation test. We reasoned that space-forms from SSS should follow some topological rules reflecting the ordinality properties of the stimulus. A direct analogy comes from cartography, given spatial continuity underlying these representations, they need to

Figure 4

Example self-intersections test

Note. Average of the total number of self-intersections across repetitions and categorie. In this example participant, the score is *1.44*

reflect some topological rules too.

Similarly to the self-intersection test, topological simplicity evaluates whether a geometric object has a simple structure without self-intersections or self-tangencies. According to the OGC Simple Features Specification (Herring, 2010), a polygon boundary is considered simple if it does not pass through the same point more than once (*PostGIS 3.6.2dev Manual*, n.d.).

Topological validity tests if a polygons is well-formed and valid according to the Open Geospatial Consortium (OGC) Simple Features Specification (Herring, 2010). A polygon is considered topologically valid if it satisfies the following criteria: (1) if polygon rings are simple (i.e., they do not touch or self-intersect), (2) boundary rings to not cross (3) boundary rings may only touch points tangentially (4) rings that define holes are contained within the exterior ring (5) the polygon rings must not splits the polygon (*PostGIS 3.6.2dev Manual*, n.d.). Note that simplicity is a condition for validity: a polygon can be simple but still invalid (e.g., if an interior ring extends outside the exterior ring).

Topological simplicity and validity evaluates each space form with a binary outcome (0 = is simple/valid, 1 = complex/invalid).Topological validity and simplicity was tested across individual 3 categories (weekdays, month and numbers) and 3 repetitions, hence 9 space-forms or polygons.

For the example of a participant, having all weekday's space-froms polygons being invalid across repetitions (i.e. 1,1,1), all month's being invalid (i.e. 0,0,0), and numbers invalid in one of the three repetitions (0,1,0), then the topological validity score would be 4/9 ≈ 0.44. Hence, validity and simplicity of each participants span from 1 (all 9 are valid/simple) to 0 (none of the 9 forms are valid/simple).

Compactness score. We then implemented a compactness score of the space-formed polygons. We used the score developed to quantify the gerrymandering of political district Polsby and Popper (1991). In the context of geo-politicity, gerrymandering is a politically motivated strategy of redefining electoral district's boundaries to advantage a political party. As a consequences, the compactness of districts redefined by political motivations are less likely to be compact than if they would follow geomorphological properties. We expected SSS space-form polygons to be more compact for SSS than controls since those polygons are grounded in a perceptual space.

The compactness score is calculated as the ratio of the area and perimeter of a polygon. Formally, the Polsby–Popper score is calculated from the polygon's area and perimeter Equation 3:

$$P - P\ score = \frac{4\pi \cdot Area}{Perimeter^2} \tag{3}$$

Hence, this score ranges from 1 (maximally compact) to 0. For space-forms that have an area of 0 (for example when all the responses are in the same position), the score could not be computed, we therefore re-defined those polygon's p-p score to 1.

For example, for a polygon overlapping to a perfect circle, the polsby-popper score would be 1. Indeed, the maximal participant's scores displays circular space-form, see Figure 3.

Permutation-based feature extraction. If synesthetic forms are truly consistent, they should remain stable independently of the experimental order in which the stimulus is presented, see Figure 1. Hence, we re-calculated the three most per formant tests (i.e. leading to higher Area under the Curve) by permuting the repetition order within each condition (see Section 2.3.2). We predicted that this permuted-based consistency would yield to better classification performance (higher AUC values) compared to non-permuted methods.

Each permutation was obtained by shuffling the repetition order or label between stimulus presentation (i.e. Monday (repetition 2) → Tuesday (repetition 3) → Wednesday (repetition 1), ect.). Then we extracted the geometrical feature from x,y coordinates of stimuli from the same categories in non-chronological order. We repeated 500 permutations per participants and averaged the permuted distributions to obtain a single permutation score per participant. The permutation scores were calculated for topological validity and Polsby–Popper score.

For example, segments for weekdays were constructed with the x,y coordinates randomly permuted across repetition order, i.e. Monday (repetition 1), Tuesday (repetition 3), Wednesday (repetition 2), ect., see rightmost panel of Figure 1.

2.3.3 Receiver Operating Characteristic analyses

Each of the previous method's performance in classifying SSS from controls was compared with Receiver Operating Characteristics (ROC) analyses. Classification performance was quantified using the area under the curve (AUC), with higher AUC values indicating better discrimination between SSS and control groups. The optimal cutoffs were calculated using Youden's index (Youden, 1950). Discriminant Power (DP) was used as an additional estimate of discrimination performance according to the optimal cutoff (Equation 4). DP around 1 being interpreted as inefficient discrimination.

$$DP = \frac{\sqrt{3}}{\pi}(\log(\frac{sensitivity}{1sensitivity}) + \log(\frac{specificity}{1specificity})) \tag{4}$$

The statistical comparison of ROC curves between the methods was operated testing the difference of AUC between pairs of curves using bootstrapping method (1000 iterations) since it allows to compare methods with different directions. Each bootstrap results in the subtraction of the AUC between the two ROC. Then the each of the AUC difference's (D) are divided by the standard deviation of those difference and compared to a normal distribution (Equation 5).

$$D = \frac{AUC_1 - AUC_2}{SD} \tag{5}$$

Therefore we statistically compared all the AUC of each method as the difference to the highest AUC method.

3 Results - phase I

Regarding the reproduced consistency test, the available datasets resulted in descriptively different averages for SSS, see Table 3. The triangle area (in % screen size) for SSS was surprisingly larger in the present aggregated dataset 0.26% (*SD = 0.52%*) than in previous studies studies: 0.14% (*SD = 0.13%*) in Rothen et al. (2016) and 0.14% (*SD = 0.17%*) in Ward et al. (2018). Descriptives break down by dataset (see Appendix B) suggest that the dataset with the largest number of participants (Ward, 2022) is driving this difference (i.e. with an area of 0.26%, see Table B1). It is important to note that since each method uses different metrics, these cannot directly be compared to each others. Hence the rightmost columns of Table 3 also reports standardozed z-score descriptives allowing comparison across different metrics. Furthermore note some the directionality differs between the methods (i.e. the area method is expected to be smaller for SSS in contrast to validity methods that are expected be larger for SSS).

In sum, the descriptives stress out the variability across datasets and the need for additional analyses from phase II. It also stresses out the use of methods that are either standardized or that do not rely on metrics that depend on the experimental settings.

3.1 ROC analyses

The results of the ROC analyses are summarized in Table 4 and Figure 5. The consistency methods are sorted by highest Area Under the Curve (AUC). We applied statistical comparisons of all the ROC curves by pairs. We compared the permuted validity method that

Table 3

Descriptive values of each test per group

Feature	Control	SSS	Control [zs]	SSS [zs]
Area [screen size %]	0.46 (0.87)	0.26 (0.52)	0.17 (1.25)	-0.12 (0.75)
Area [z-score]	0.26 (0.32)	0.08 (0.14)	0.42 (1.29)	-0.30 (0.55)
Perimeter [screen size %]	0.03 (0.03)	0.03 (0.08)	0.03 (0.44)	-0.02 (1.26)
Perimeter [z-score]	3.30 (1.72)	1.59 (1.16)	0.60 (1.04)	-0.44 (0.70)
Segment self-intersections [n/9]	9.56 (11.31)	1.87 (5.42)	0.48 (1.23)	-0.35 (0.59)
Area of the polygon [z-score]	1.00 (0.99)	1.91 (1.31)	-0.42 (0.79)	0.30 (1.03)
Perimeter of the polygon [z-score]	9.49 (3.44)	8.39 (2.48)	0.21 (1.16)	-0.16 (0.83)
Simple topology [boolean/9]	0.22 (0.23)	0.40 (0.27)	-0.39 (0.85)	0.28 (1.01)
Valid structure [boolean/9]	0.13 (0.19)	0.39 (0.28)	-0.54 (0.69)	0.39 (1.01)
Permuted valid structure [boolean/9]	0.12 (0.16)	0.36 (0.25)	-0.57 (0.64)	0.41 (1.01)
Polsby–Popper [score/9]	0.11 (0.13)	0.27 (0.19)	-0.52 (0.69)	0.38 (1.02)
Permuted P-P [score/9]	0.11 (0.17)	0.41 (0.33)	-0.56 (0.56)	0.40 (1.06)

Note: mean (standard deviation) of all the consistency tests. The two rightmost columns display the same descriptives in z-score transformed values, enhabling comparison across different metrics. Z-scores were calculated for all participants, separately for each test.

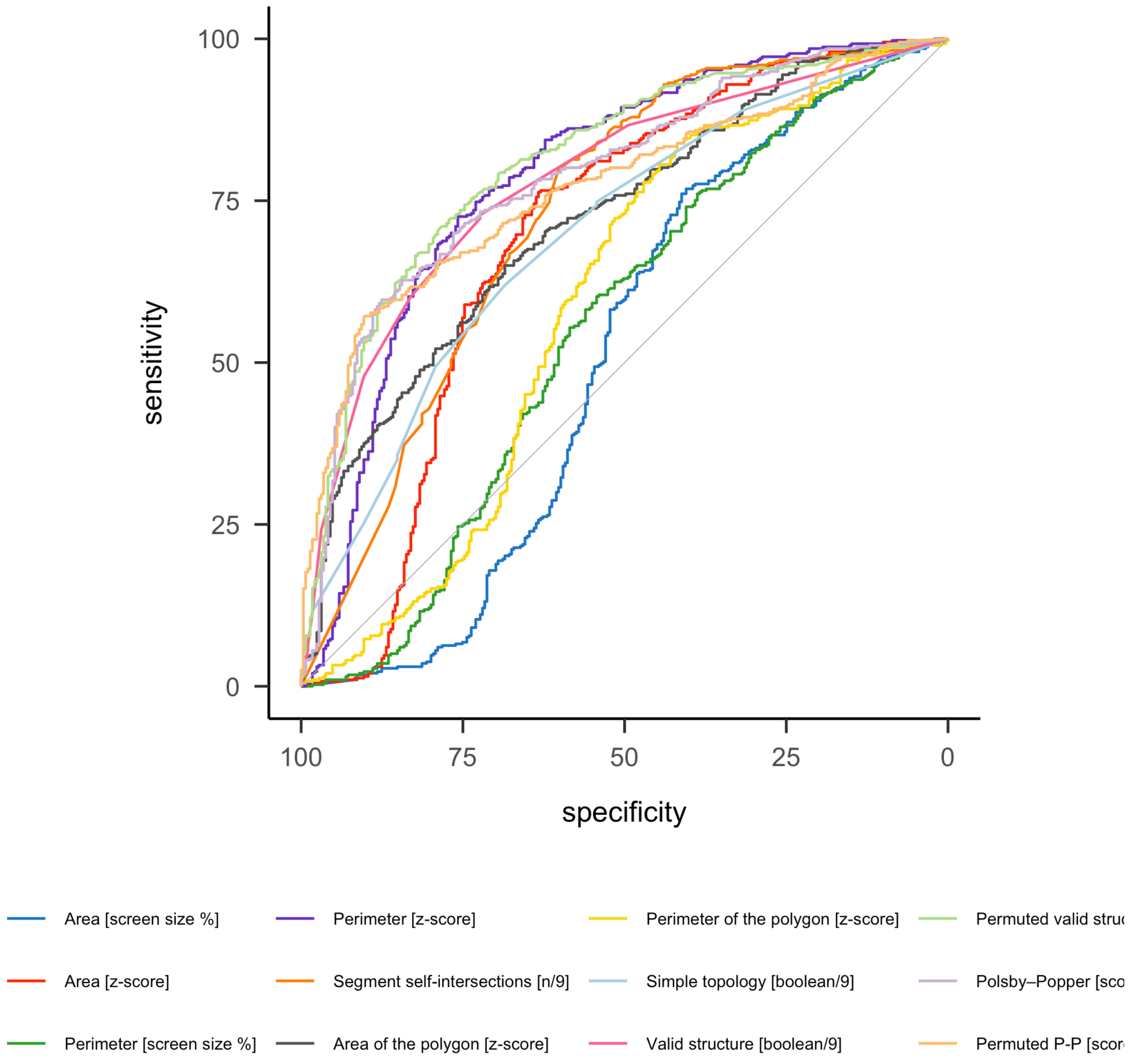
lead to the highest AUC as described in Section 2.3.3. All p-values are adjusted with false discovery rate.

Permuted validity and the standardized perimeter led to the highest AUC and were statistically the same ($D = 1.4$, $p = n.s.$). The Polsby-Popper method, did statistically differ from the permuted validity ($D = 2.03$, $p < .05$), while counter-intuively having descriptively larger AUC than the standardized perimeter test, see Table 4. All the other tests led to significantly smaller AUC than permuted validity (i.e. all $D > 3.18$, $p < .01$).

We conducted further analyses to compare the sensitivity and specificity of the Polsby-Popper to the permuted validity that resulted in no significant differences. Permuted validity and Polsby-Popper method did not differ in specificity (at a sensitivity of 80%, $Z = 0$,

Figure 5

Receiver Operating Characteristic (ROC) curves of all tests



Note. Grey line indicates chance level

Table 4

ROC analyses of all tests

Feature	AUC	DP	threshold	sensitivity	specificity
Permuted valid structure [boolean/9]	81.16	2.13	0.17	70.78	78.55
Polsby–Popper [score/9]	78.96	2.29	0.19	59.70	87.54
Perimeter [z-score]	78.80	2.06	1.73	72.54	75.78
Valid structure [boolean/9]	78.12	1.90	0.14	72.80	72.32
Permuted P-P [score/9]	77.14	2.46	0.20	57.18	90.31
Segment self-intersections [n/9]	72.93	1.75	1.17	79.85	60.21
Area of the polygon [z-score]	72.03	1.36	1.29	64.99	68.51
Simple topology [boolean/9]	69.74	1.24	0.28	61.96	68.51
Area [z-score]	69.27	1.68	0.08	76.32	63.32
Perimeter of the polygon [z-score]	59.12	1.29	10.46	83.88	41.87
Perimeter [screen size %]	55.34	0.68	0.03	76.07	38.75
Area [screen size %]	50.87	0.79	0.21	76.83	40.48

AUC = Area Under the Curve. DP = Discrimination Power.

p = *n.s.*). However, comparing specificity to the standardized perimeter resulted in higher specificity for the Polsby-Popper method (at a sensitivity of 80 %, $D = 2.04$, $p < .05$). For sensitivity, the Polsby-popper method did not differ neither from permuted validity (at a specificity of 80%, $D = 1.06$, p = *n.s.*) nor from perimeter (at a specificity of 80%, $D = 0.98$, p = *n.s.*). See also Figure A1 for the density plot's distribution across self-reported SSS and controls.

The results also show that the permutations significantly improved the AUC for topological validity ($D = 3.18$, $p < .01$). Post-hoc comparison of the Polsby-Popper method to it's permuted counterpart indicate (one-sided bootstrap comparison of the AUC: $D = -1.78$, $p < .05$). In sum, the permutations only selectively improved classification performances of topological validity but not the compactness Polsby-Popper score.

In conclusion for the statistical analyses, while the permuted validity and standardized perimeter lead to overall higher AUC than the Polsby-Popper, the later resulted in significantly higher sensitivity than perimeter (i.e. better probability of detecting SSS) than standardized perimeter, see Table 4. Hence the permuted validity, standardized perimeter and Polsby-Popper were the most optimal consistency methods for the aggregated dataset.

The cut-off leading to the most optimal classification for the permuted validity score is: 0.17 corresponding to 1.53 valid of 9 forms. The cut-off leading to the most optimal classification for standardized perimeter is: 1.73 of individual *z-scores*. The cut-off that leads to the most optimal classification with the compactness score obtain from the Polsby-Popper method is: 0.19, the compactness score is calculated on the space-form polygon's area and perimeter using Equation 3.

In sum, the novel permuted validity method and Polsby-Popper compactness score resulted in comparable classification performances between controls and SSS to the standardized coordinate's perimeter method.

3.2 Further analyses

We conducted further analyses by sub-sampling the dataset source and questionnaire percentiles, see appendixes. The first aim of these analyses was to test the stability of the classifications from different methods across experiments (i.e. datasets) and sample characteristics (i.e. questionnaire scores). The second aim was to describe the false positves and negatives of each tests to check for some biases towards some space-forms or responses in the same position. We also used a correlational approach to test for correlations between method's metrics and questionnaires.

We sub-sampled the dataset source for the five most optimal tests (i.e. higher AUC) and recalculated AUC, sensitivity and sensibility. We found similar values for the different datasets, see Figure B1. The density plots also seem to support bimodal distributions by datasets Figure B2.

Including only the data by Jamie Ward, we sub-sampled the participants depending on the percentiles of the questionnaire scores , taking the 10% highest questionnaire scores against the 10 % lowest (i.e. 90-100 %). Here also most of the values remain relatively stable independently from the questionnaire distribution of the samples, see Figure C1. Another important point addressed in the literature about consistency test is circularity.

The circularity is given in that if consistency tests are used to classify SSS and controls, then those groups will by definition differ in consistency (Root et al., 2025). Hence, we made a correlation matrix with the questionnaire scores and all the tests, see Appendix. The two highest correlations are indeed between the questionnaire score and perimeter (r = .58) and permuted validity (r = .50), see Figure D1.

We developed an averaging approach to visualize the space-forms across all participants by groups (SSS vs controls) and categories (month, number and weekday). The figure in the appendix (Figure E1), shows more circular patterns for months and more linear horizontal and vertical for numbers. We used the averaging approach to visually qualify the false negatives (Controls classified as SSS) and positives (SSS classified as controls) of each consistency test. Descriptively, the perimeter test seems to classify linear responses for numbers as controls compared to the other methods.

Finally, some of the participants might have been originally mis-attributed to control or SSS or there might be consistent controls and inconsistent synesthetes (Eagleman et al., 2007; see Root et al., 2025), see Figure F1. We found ~6% of the full sample's synesthetes to be consistently classified as controls (i.e. consistent false negatives) by the five methods with the highest AUC, see Figure F2. ~3 % controls are consistently classified as SSS (i.e. consistent false positives) Figure F3. Qualitatively, the consistent false negative seem not to be like

In sum, these further analyses evidence first, some variability across datasets. Second, the main consistency tests seems to stably classify more extreme groups of SSS and controls than less extremes ones (based on questionnaire score distribution). Third, qualitatively less false-positives (SSS classified as controls) for the novel space-form methods (i.e. validity and compactness) than the old perimeter method. Fourth, we identified a group within the controls who systematically pass all the consistency tests and a group within the SSS who systematically fail all consistency test, accounting for about 9% of all participants.

4 Methods - phase II

Additional data will be collected in the future using the same consistency test, with the procedural exception that there will be four repetitions per stimuli instead of three (Sachdeva et al. (2024), Stage 1 in-principle acceptance). Hence for the stimulus levels tests we will compare the area and perimeter of a quadrilateral of four coordinates pairs instead of a triangle. Materials are more details on this study are pre-registered on OSF, including ample size and recruitment method (Sachdeva et al., 2024, Stage 1 in-principle acceptance) (see also https://osf.io/6wn4m).

5 Discussion - phase I

Our investigation of four datasets found two main consistency tests leading to the optimal classification of SSS from control: standardized perimeter between stimulus repetitions and permuted topological validity on the space-form level.

Further analyses show variabilities between the test's AUC exists depending on the dataset source, which could explain why we could not replicate Rothen et al. (2016)'s area but rather found the perimeter as a more optimal test. The stability across varying percentiles of the synaesthesia questionnaire's score suggest the tests work as well for more marked forms of SSS (note however that this analysis was limited to Jamie Ward's data). The standardized perimeter also leads to slightly higher correlation with the questionnaire scores (r = .57) than permuted topological validity (r = .51), see Figure D1.

In sum, while we could prove similar performances using a different approach based on the space-form level (i.e. topological validity), we found similar classification performances than using stimulus-repetition level test such as perimeter.

5.1 Limitations

Although an optimal test to classify SSS might be particularly relevant for experimental purposes, several important limitations need to be considered. First, consistency tests contain only a limited set of sequential stimuli (i.e. months, weeks and the first ten natural numbers) that could potentially elicit synaesthetic experiences. Other ordinal categories such as temperature, clock time, musical keys might be more relevant for some individuals with SSS. For numbers specifically, better consistency tests might be obtained using a larger set size (i.e. including decades and hundreds), as descriptively interesting form changes occur at different decimals in base-10 number systems (Galton, 1880).

Second, the use of diagnostic cutoffs assumes categorical distinctions between groups, but SSS may exist on a continuum and scores might be more suitable (Price & Pearson, 2013). This limitation is related to the circularity mentioned previously: diagnostic cutoffs as calculated here depend on how SSS and controls are classified in the first place (Simner, 2012). Measures of other characteristics of synaesthesia such as automaticity or visual Gabor detection might be necessary to establish external validity (Ward et al., 2018). Indeed, determining the prevalence of SSS in the general population requires definitional choices that might be based on more conservative or lenient criteria (Brang et al., 2010; Jonas & Price, 2014; Sagiv et al., 2006; Ward et al., 2018). These difficulties are reflected in the span of current prevalence estimates of SSS: 4.4 % (Brang et al., 2013), 8.1 % (Ward et al., 2018) and 14 % (Seron et al., 1992).

Third, as shown in Figure B1, we find that the optimal criteria can vary across datasets. In other word, we obtain different AUC depending on the datasets. These differences might be explained by different sampling methods biases, for example (Van Petersen et al., 2020) recruited SSS from over one hundred candidates to maximise participant reporting synesthetic experiences.

5.2 Topological validity

Surprisingly, permuted topological validity of the spatial-forms led to similar results than the perimeter test. This novel result might be informative about how SSS map ordinal stimuli in space. Indeed, it seems the patterns of spatial-forms follow topological rules analogous to geographical and geometry-based space structures. Analogies between maps and neuroscience have a long history (i.e. retinotopy, sonotopy or somatotopy Eagleman & Goodale, 2009). Therefore spatial-forms might originate from the mapping between ordinal or sequential stimuli on idiosyncratic visuo-spatial abilities following topological rules. One of the advantages of using topological validity is that it also classifies responses with the same coordinates as invalid and hence inconsistent, while using the area and perimeter metrics they would qualify as highly consistent responses.

5.3 Intermediary Conclusion

The present study compared traditional stimulus-level consistency tests with novel form-level geometric tests for detecting SSS. We found optimal classification performance from permuted topological validity and standardized perimeter between repetitions. Despite not yielding to better classification than stimulus level methods, this new method is theoretically meaningful since genuine synesthetes should produce more compact and topologically valid structures compared to controls. Similarly, the perimeter test captures the stability of the overall spatial configuration across repetitions, which should remain consistent for true SSS individuals regardless of presentation order.

In the future study, we will attempt at validating these criteria on a yet-to be acquired dataset.

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6.0.1 Appendix

The additional analyses in this appendix attempt to address several concerns. Regarding the distribution's of the scores from the different methods, i.e. whether it is bimodal, see Appendix A. Then to visualize the stability of the ROC analyses across the datasets, i.e. whether AUC, DP, sensitivity and specificity vary across the four different datasets, see Appendix B.

To address circularity concerns that we might find methods that best classifies synesthetes and controls only from the self-reported groups, we attempted two additional approaches: first, testing sub-samples based on percentiles of the questionnaire score sample distribution (see Appendix C). The rationale is that a good consistency test should have similar discriminative performance for strong SSS and weaker SSS. To do this, we re-sampled the SSS and controls based on their questionnaire distribution by percentiles (i.e., comparing 10% highest questionnaire scores vs. 10% lowest, then 20% vs. 20%, etc.). In addition, we correlated the questionnaire scores with the results from different features extracted from the consistency test, see Appendix D.

We also wanted to visually characterize whether certain patterns occur more frequently for some categories than others (i.e. circular pattern for months and linear for numbers). Appendix E shows the average forms for each categories. Finally, we wanted to visualize how each participant is classified by each test. From this we found a group of Synaesthetic who consistently are classified as controls by different methods as well as a group of control who are consistently classified as synaesthetes, see Appendix F.

The following R packages were used for the data visualization in the appendix: 'ggridges' v. 0.5.7 (Wilke, 2025b), 'cowplot' v. 1.2.0 (Wilke, 2025a), 'ggalluvial' v. 0.12.5 (Brunson & Read, 2023), 'corrplot' v. 0.95 (Wei & Simko, 2024)

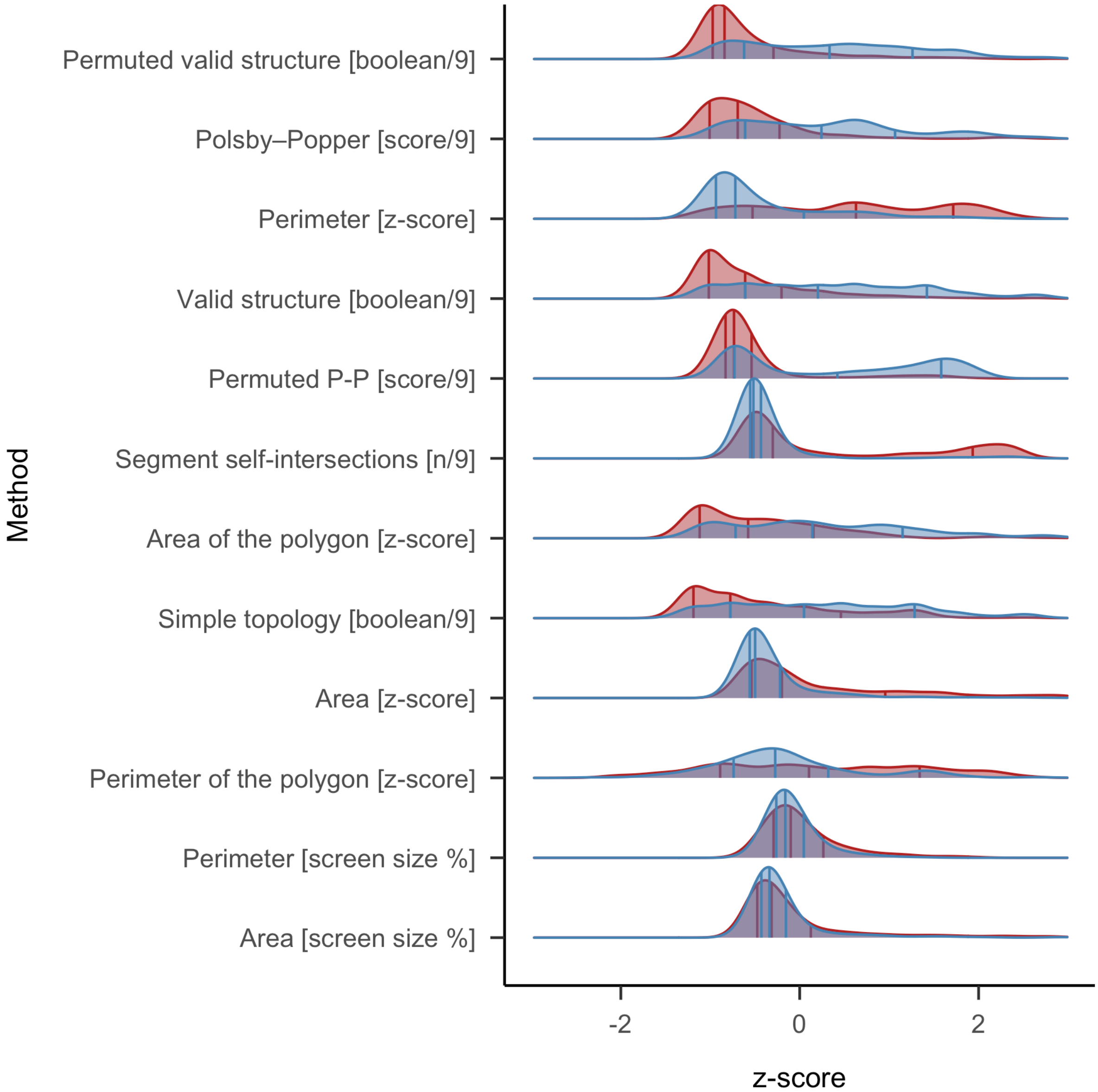
Appendix A

Test's distributions

Regarding the distributions of each tests across the SSS vs. controls, Figure A1 we compared the density plots of each standardized criteria (in order to make them comparable) which visually suggests bimodal distribution.

Figure A1

Density plots of all the consistency tests comparing SSS and controls



Note. Red = Classified as Synesthetes, Blue = as controls.

All method's score have been standardized (*z-score*). x axis trimmed between -3 and 3 z-scores.

Vertical line represent respectively 20, 50 (median) and 80 percentiles of participant's distributions by self-reported SSS or Control.

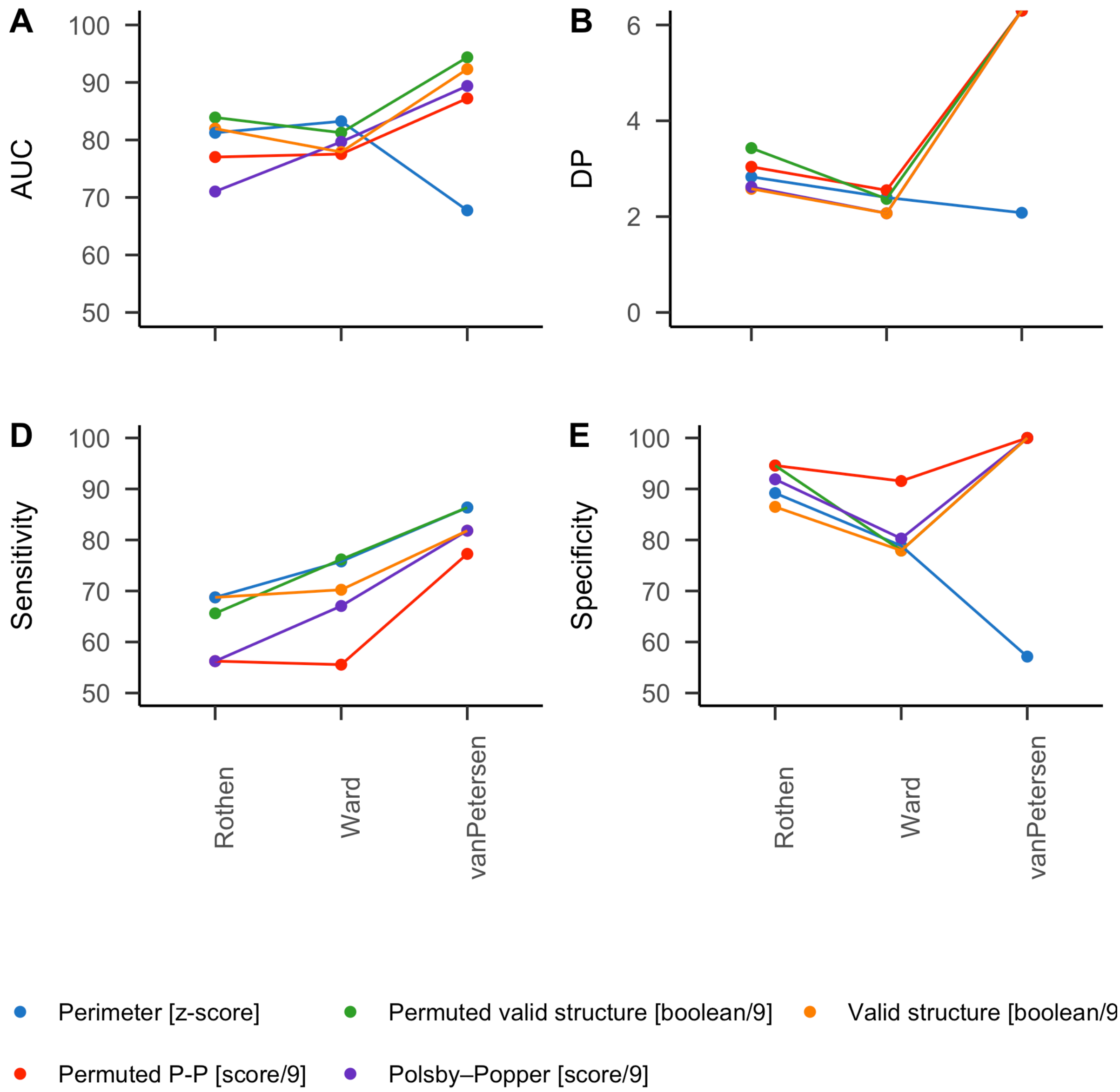
Appendix B

ROC analysis by dataset

Here we compare the ROC analyses results for each data sample of the three methods with the best AUC, see Figure B1 and Table B1. Descriptively, we also looked a the distributions of each methods across SSS and controls Figure B2. Note that the dataset labelled as Ward 2 has more synaesthetes than controls (5:1 ratio). Note that some of the tests led to 100 % Specificity in Van Petersen et al. (2020) dataset.

Figure B1

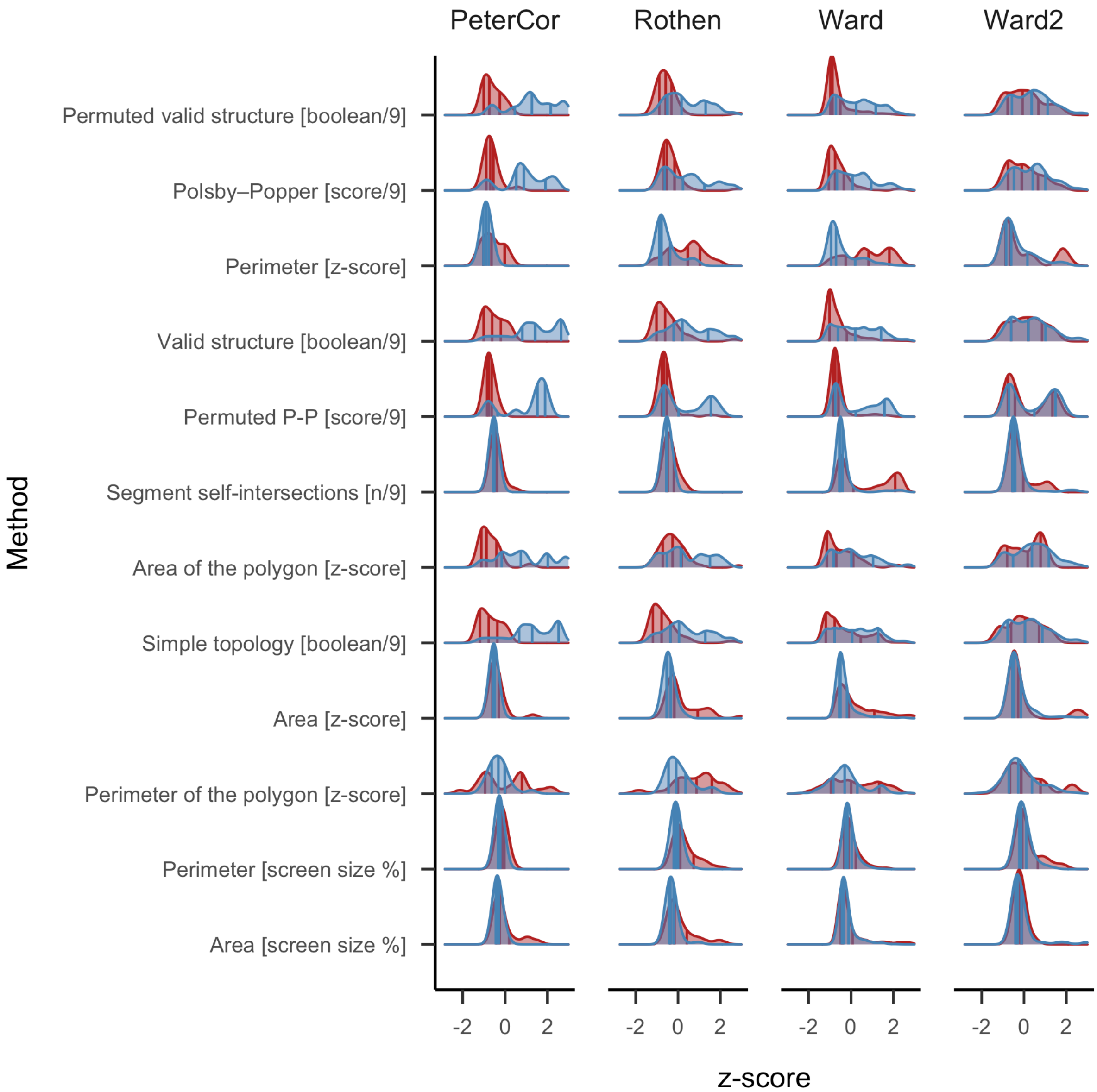
Lineplots of AUC and DP by data source



Note. AUC = Area Under the Curve, DP = Discrimination Power

Figure B2

Density plots of all the consistency tests comparing SSS and controls by dataset



Note. Red = Classified as Synesthetes, Blue = as controls.

All method’s score have been standardized (*z-score*). x axis treamed between -3 and 3 z-scores.

Vertical line indicate respectively 20, 50 (median) and 80 percentiles of each distributions.

Table B1

Average feature for Synesthetes by dataset

	Control				SSS			
Feature	PeterCor	Rothen	Ward	Ward2	PeterCor	Rothen	Ward	Ward2
Permuted valid structure [boolean/9]	0.11	0.13	0.11	0.27	0.59	0.40	0.33	0.35
Polsby–Popper [score/9]	0.08	0.14	0.10	0.21	0.46	0.29	0.26	0.27
Perimeter [z-score]	1.33	2.91	3.66	2.25	0.84	1.41	1.62	1.75
Valid structure [boolean/9]	0.13	0.14	0.11	0.31	0.66	0.43	0.36	0.37
Permuted P-P [score/9]	0.06	0.11	0.10	0.33	0.61	0.40	0.40	0.41
Segment self-intersections [n/9]	1.41	1.78	12.26	3.02	0.22	0.26	2.23	1.85
Area of the polygon [z-score]	0.63	1.27	0.94	1.61	2.75	2.15	1.75	2.06
Simple topology [boolean/9]	0.13	0.14	0.23	0.32	0.66	0.43	0.38	0.38
Area [z-score]	0.06	0.23	0.29	0.19	0.02	0.05	0.08	0.09
Perimeter of the polygon [z-score]	8.87	11.21	9.27	9.21	7.98	9.06	8.29	8.50
Perimeter [screen size %]	0.02	0.05	0.03	0.04	0.01	0.03	0.03	0.03
Area [screen size %]	0.31	0.89	0.37	0.80	0.10	0.14	0.26	0.34

Table B2

Average zs feature for Synesthetes by dataset

	Control				SSS			
Feature	PeterCor	Rothen	Ward	Ward2	PeterCor	Rothen	Ward	Ward2
Permuted valid structure [boolean/9]	-0.61	-0.52	-0.62	0.05	1.36	0.60	0.32	0.39
Polsby–Popper [score/9]	-0.66	-0.36	-0.57	0.01	1.39	0.47	0.28	0.37
Perimeter [z-score]	-0.59	0.36	0.81	-0.03	-0.89	-0.54	-0.42	-0.34
Valid structure [boolean/9]	-0.53	-0.50	-0.60	0.13	1.39	0.57	0.31	0.33
Permuted P-P [score/9]	-0.73	-0.58	-0.59	0.13	1.04	0.38	0.35	0.40
Segment self-intersections [n/9]	-0.40	-0.36	0.78	-0.23	-0.53	-0.53	-0.31	-0.35
Area of the polygon [z-score]	-0.71	-0.20	-0.47	0.06	0.97	0.49	0.18	0.42
Simple topology [boolean/9]	-0.70	-0.66	-0.34	0.00	1.26	0.42	0.20	0.22
Area [z-score]	-0.35	0.29	0.54	0.14	-0.53	-0.42	-0.29	-0.25
Perimeter of the polygon [z-score]	0.01	0.79	0.14	0.12	-0.29	0.07	-0.19	-0.12
Perimeter [screen size %]	-0.13	0.32	-0.02	0.19	-0.27	-0.08	0.01	-0.03
Area [screen size %]	-0.04	0.79	0.04	0.65	-0.35	-0.29	-0.12	0.00

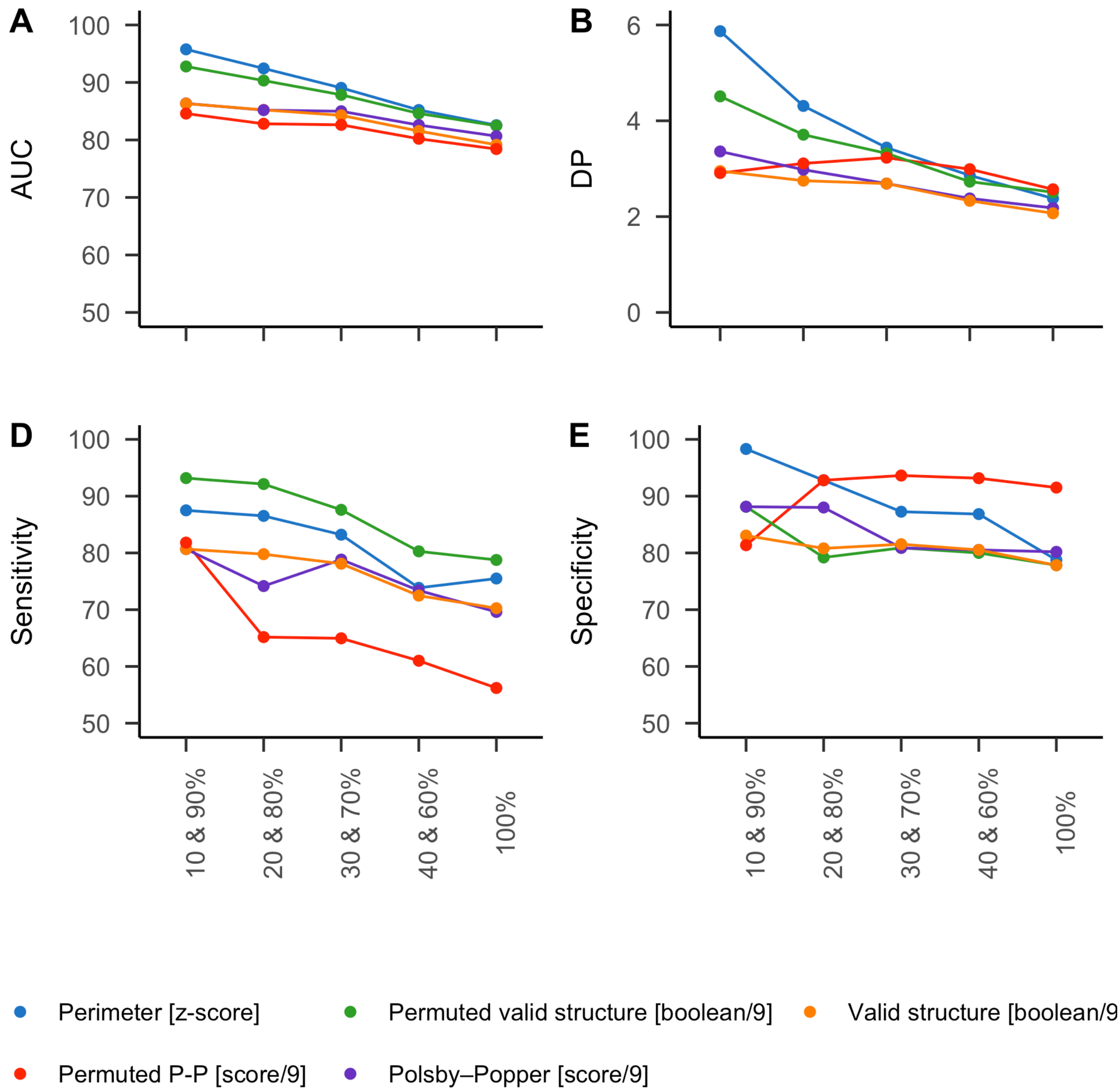
Appendix C

ROC Analysis Sub-sampled data by questionnaire quantiles (20% steps)

We compared the data sampled by the questionnaire score. Based on the distribution of the questionnaire score, we sampled the 10 % with the lowest and 10 % with the highest scores. Those are then compared with the 20 and 20 % and so on until 40 and 40 %. The rationale of this procedure is that AUC, sensitivity and specificity should remain stable across percentiles for a feature to be valid, see Figure C1. In other words the ROC should remain unchanged if we take extreme groups compared to less extreme ones.

Figure C1

Lineplots of AUC and DP by questionnaire percentile



Note. Each point is an increasing percentiles. AUC = Area Under the Curve. DP = Discrimination Power. Specitficites of some tests with van Petersen data are 100%, so the DP are nor computable nor interpretable

Appendix D

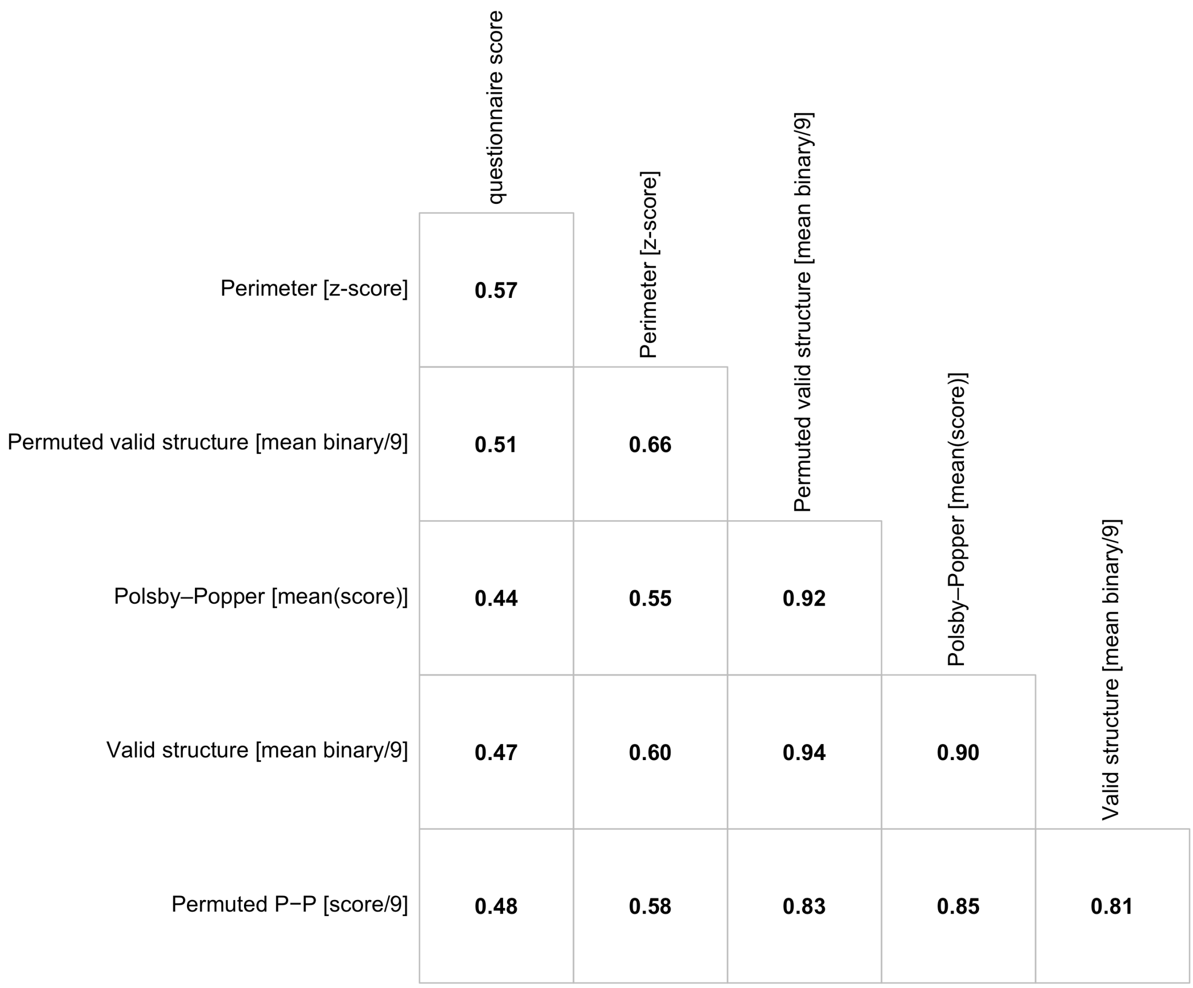
Correlation with self-report

The best criterion should also best correlate with SSS self-reported questionnaire score.

Works only with Ward's aggregated data, see Figure D1.

Figure D1

Correlation with self-reported questionnaire



Note. Only data from Ward is included hereh

Appendix E

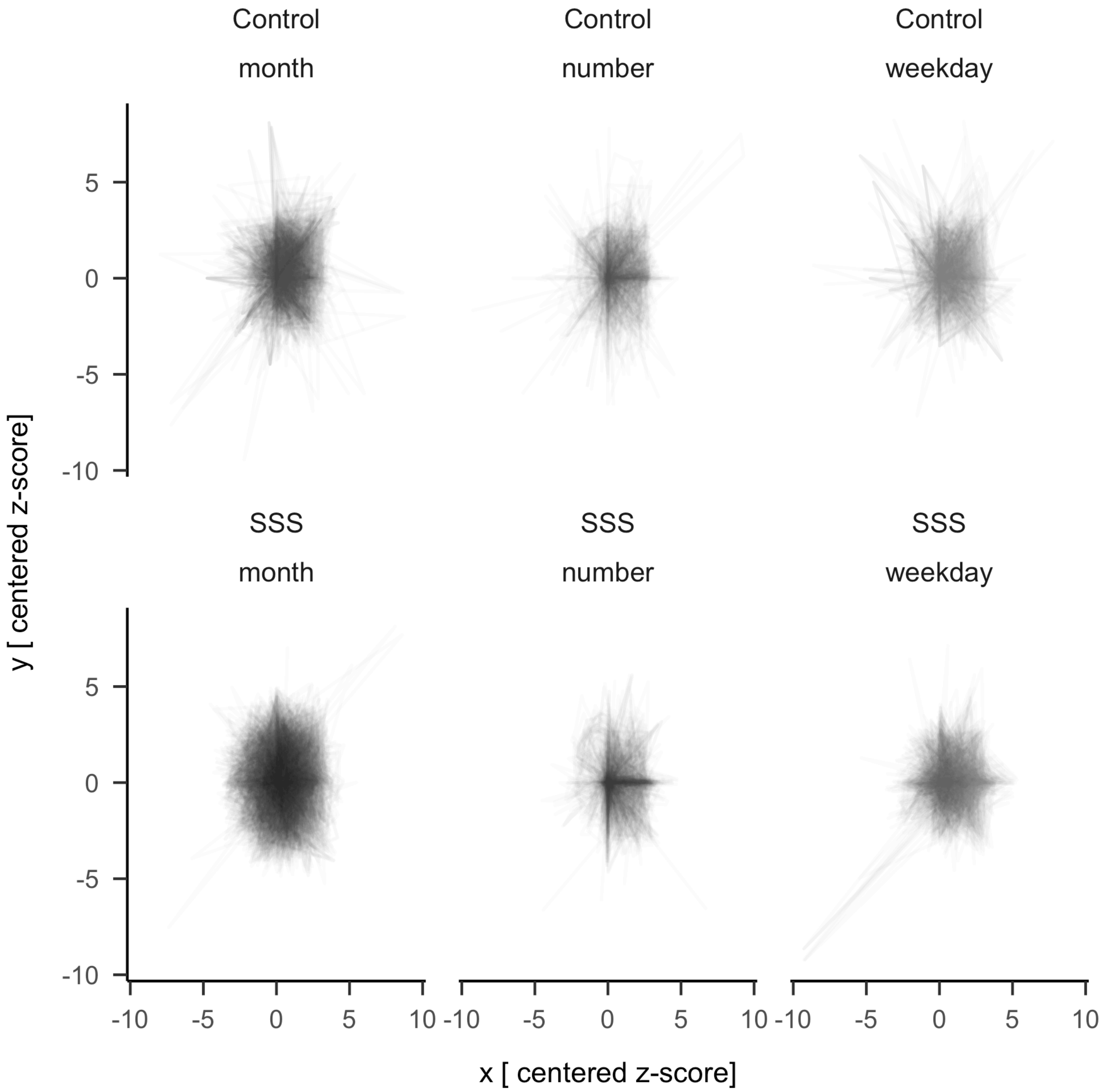
Centred average forms

In this additional analyses we attempted at visualizing the average for each categories to see whether a pattern could be discerned on a qualitative level. To avoid negative coordinates, we shifted all the z-scores from the whole dataset minimum for the x and y coordinates. Then we anchored to the coordinate 0,0 the stimulus “1” for numbers, “Monday” for weekdays and “January” for months. We then plotted all the space forms with transparency, see Figure E1.

Descriptively it is possible to see more circular pattern for months and horizontal and vertical patterns for numbers. Also the averaged forms appear to be more clearly defined in the synaesthetes than control.

Figure E1

Average Centered space forms visualized



Appendix F

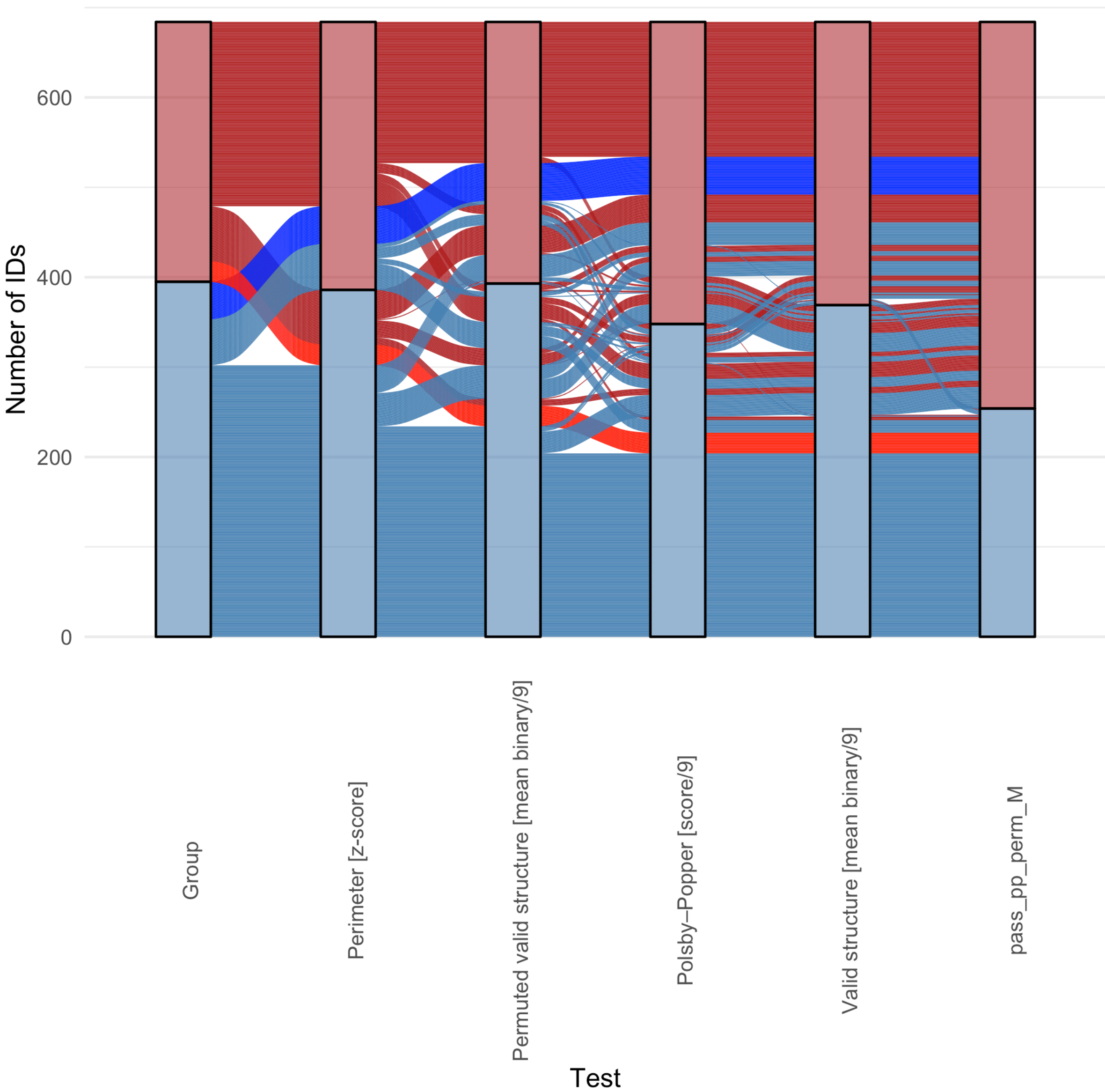
Alluvial plot

Are false positives and negatives share some characteristics? To answer this question qualitatively we identified two group of self-reported SSS participants who care consistently classified as controls by the five methods with the highest AUC (i.e. consistent false negatives), see dark blue lines in Figure F1. On the other side we identified a group of controls who are consistently classified as SSS by the five methods with the highest AUC, see dark red lines in Figure F1.

On one side 6.12 % of all participants, that is 42 Synaesthetes where consistently classified as controls by all five tests, see Figure F2. On the other side 3.35 % of all participants, 23 controls where consistently classified as SSS by all five tests, see Figure F3. Given the different approaches of each methods, the identified participants might be on one side non-genuine synesthetes and on the other controls who have SSS but are either not aware or did not identify it as such.

Figure F1

Alluvial plot of test passing



Note. Red = Controls, Blue = Synaesthetes. Highlighted in blue are Synesthetes who are classified as control by all 5 tests. Highlighted in red are controls who are classified as synesthetes by all 5 tests.

Figure F2

Synaesthetes who consistently fail *all five tests (blue in the alluvial plot)*

Figure F3

Controls who consistently pass *all five tests (red in the alluvial plot)*

Appendix G

False positive and negative analysis

To visualize whether one of the consistency tests is biased towards the classification of one category, we selected the false positives and negatives of the three best performing consistency methods (i.e. permuted validity, polsby-popper and standardized perimeter). While allowing only for visual comparison, the

Figure G1

For permuted validity: average forms per categories of false positives and negatives

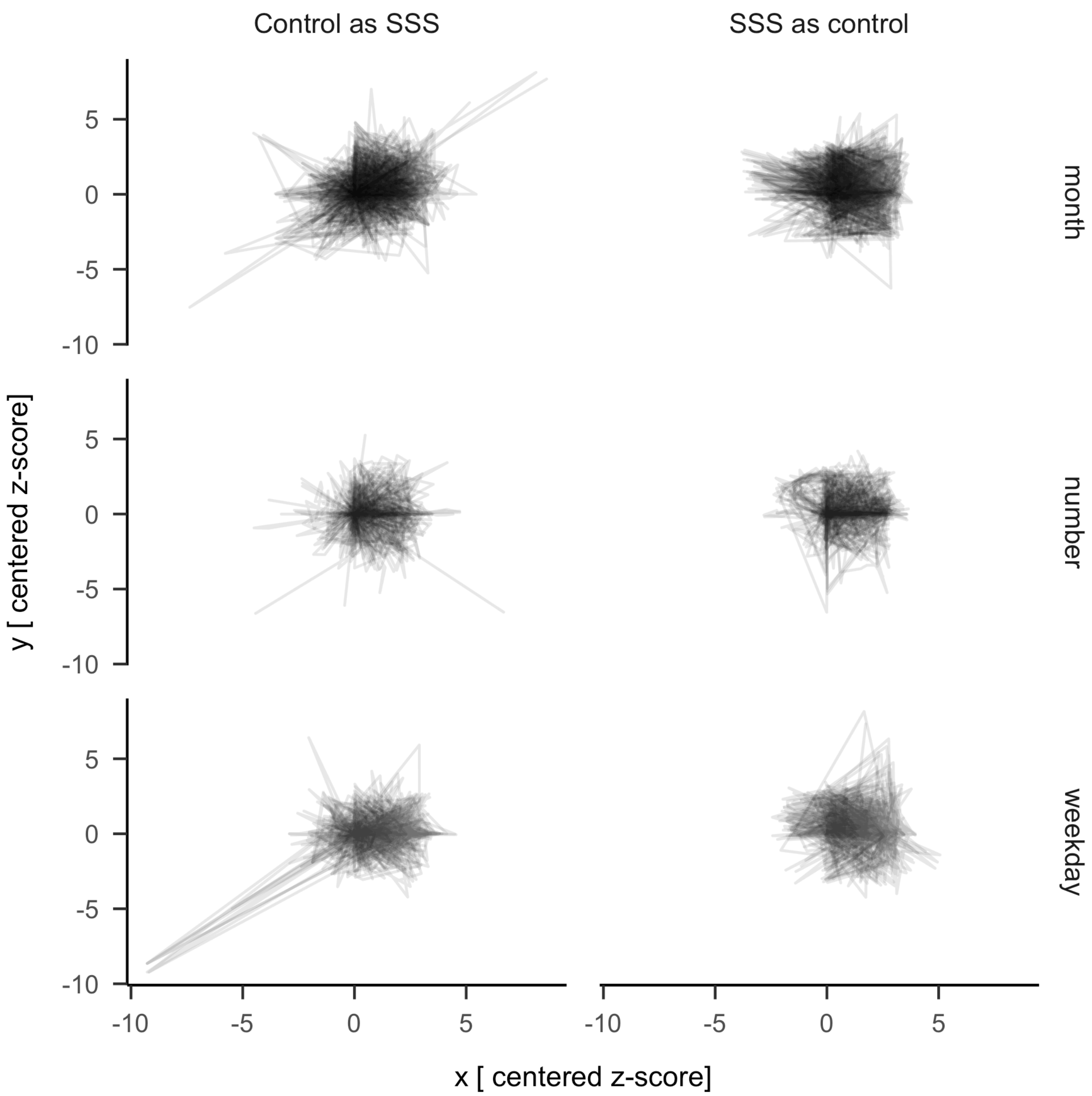


Figure G2

For Polsby-popper: average forms per categories of false positives and negatives

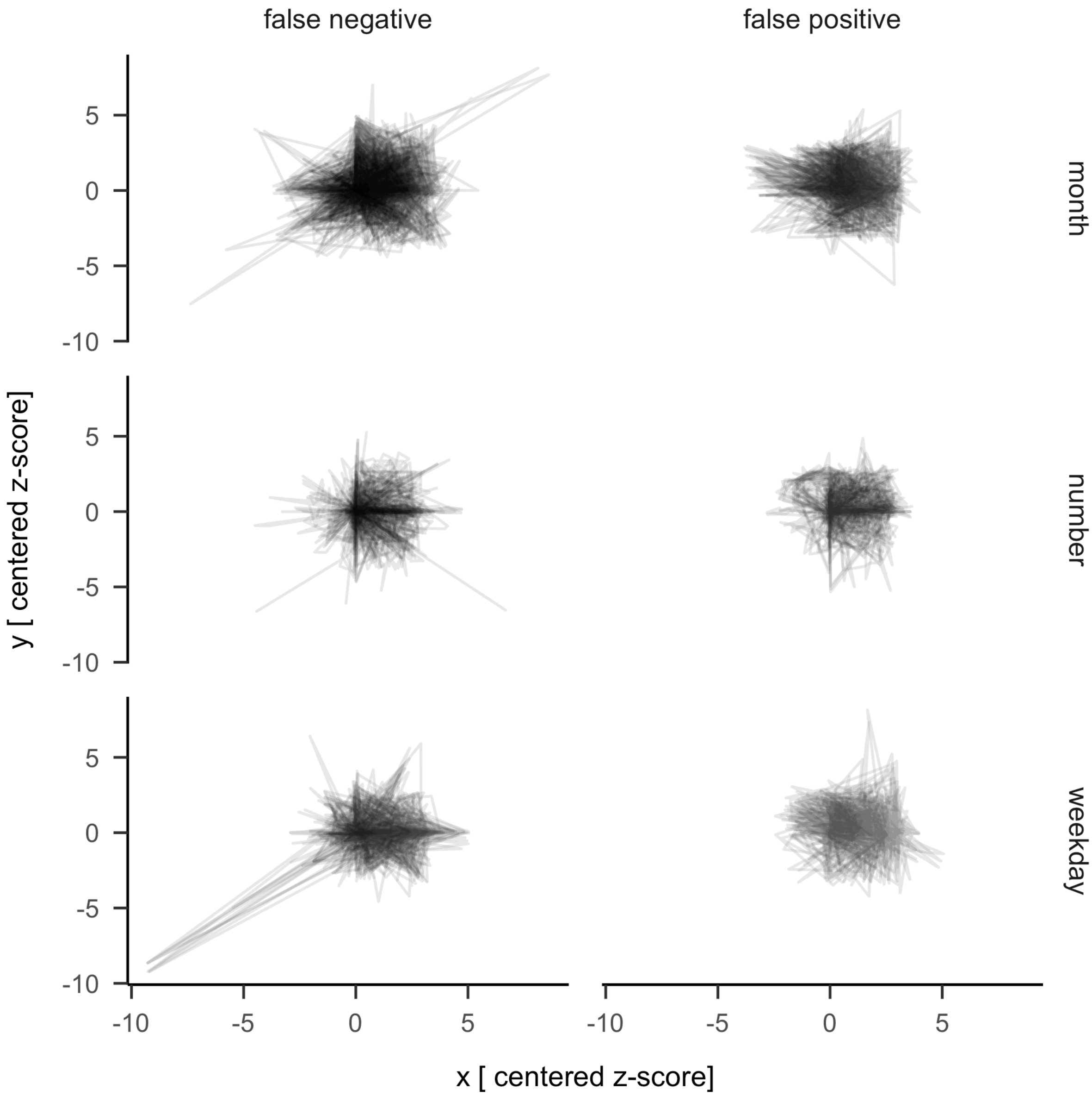
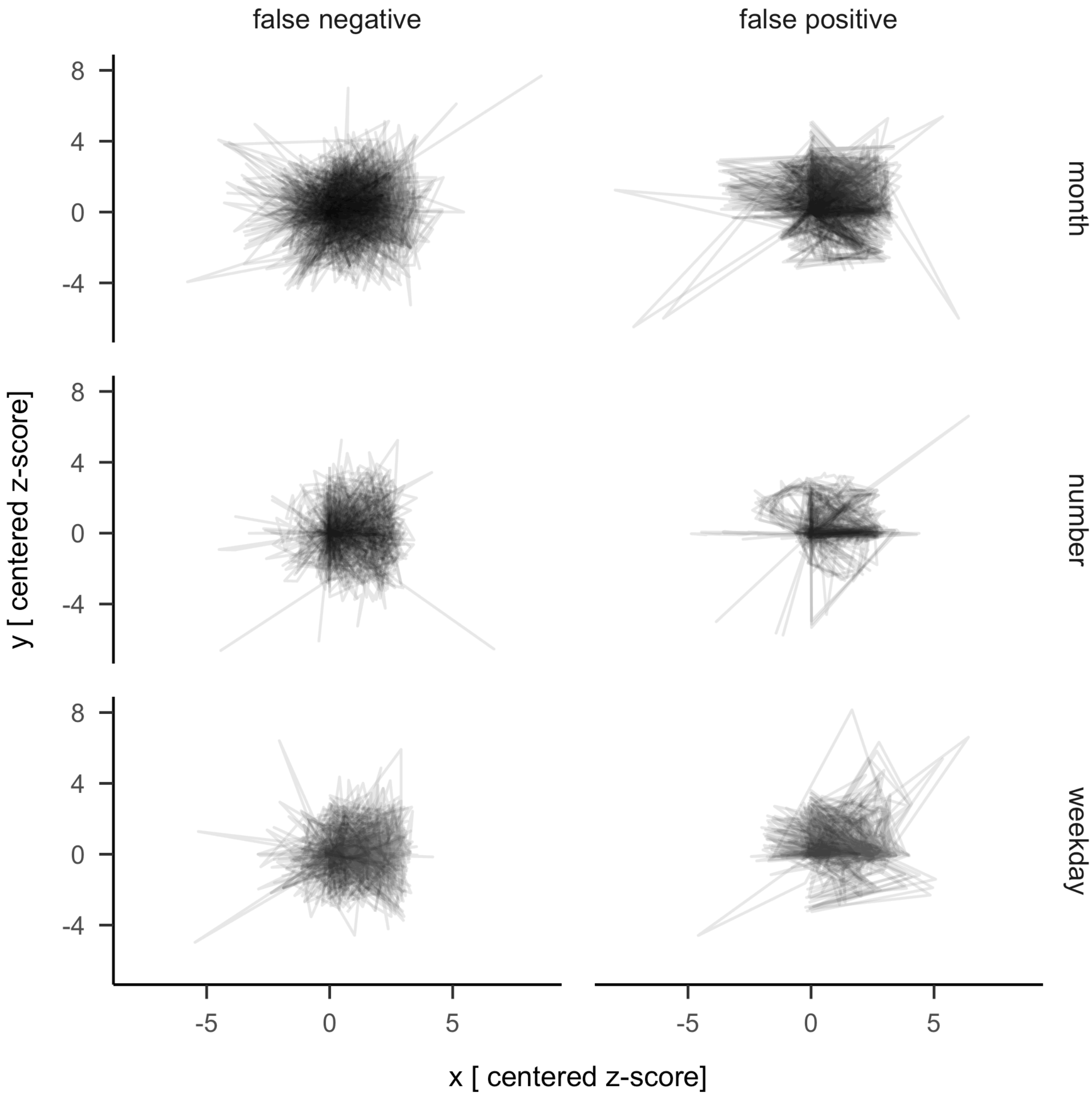


Figure G3

For perimeter: average forms per categories of false positives and negatives



Appendix H

All participants spatial forms

Here we report the code to plot each participant and categories in z-score x and y coordinates. At the top right corner, the first coured dot indicates if the participant self-reports as SSS (green) or not (red). The following dots indicates if the participant passes (green) or fails (red) the given consistency test.